

Automotive Body Microcontroller

More Function. Less Consumption



Window, Mirror



Door (Slide, rear)



Wiper



Lighting



Remote Keyless Entry



Seat



Switch, Indicator



Engine

Renesas RL78/F1x 16-bit microcontroller realize the industry's lowest power consumption and high performance for a broad range of automotive applications such as cost-effective ECUs like switches, remote keyless entry, wiper, HVAC, lighting modules and engine sub controllers. Coupled with various built-in functions in realizing functional safety, a highly reliable system can be built with Renesas RL78/F1x series.



RL78/F1x Key Features

- 16-bit device technology shrunk to 130nm
- Huge scalability with more than 120 derivatives
- New and many energy saving features integrated
- Ultra low power
 - Reduced to 50% compared to previous generation
 - Run mode: typ. 0.2mA/MHz and max. 0.4mA/MHz
- Strong performance CPU with 16 DMIPS/MHz
- Increased ambient temperature up to 150°C
- Improved Data Flash memory with minimum 100k write/erase cycles
- High integration enabling system cost reduction
 - High precision on chip oscillator (+/- 2% at -40 to 105°C) fully suitable for LIN
 - 64 MHz on chip high speed clock for dedicated peripherals